
IT EQUIPMENT COMPLAINT MANAGEMENT SYSTEM

Software Requirements Specification (SRS)

Course: Software Design & Analysis
Assignment Task 1 – Requirements Elicitation & UML

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1. Introduction

The IT Components Problem Reporting System is a platform designed to help students, teachers, and university staff report issues related to IT equipment such as computers, projectors, printers, and networking devices. In many universities, reporting IT problems is often done manually or informally, which can lead to delays, lost complaints, and inefficient maintenance processes.

This system provides a centralized digital platform where users can submit complaints about faulty IT equipment. The system records all complaints and assigns them to IT staff or technicians responsible for resolving the issues. Users can track the progress of their complaints, receive notifications about status updates, and provide feedback once the issue is resolved.

The system improves transparency, accountability, and efficiency in managing IT equipment issues within the university environment. It ensures that problems are reported quickly, assigned properly, and resolved in a timely manner.

2. Stakeholders

The following stakeholders have been identified for the IT Equipment Complaint Management System:

Stakeholder	Role	Primary Interest
User / Reporter	Faculty, student, or staff member	Report issues, track complaint status, give feedback
IT Staff	IT department members	View, assign, and resolve complaints
Technician	Repair personnel	Update repair status, add notes
Administrator	System administrator	Manage users, delete complaints, generate reports
IT Department	System owner	System uptime, security, data integrity

3. Functional Requirements

The following 20 functional requirements were elicited through stakeholder interviews, scenario walkthroughs, and analysis of existing manual processes. Each requirement is assigned a unique ID, priority level, responsible actor, and status.

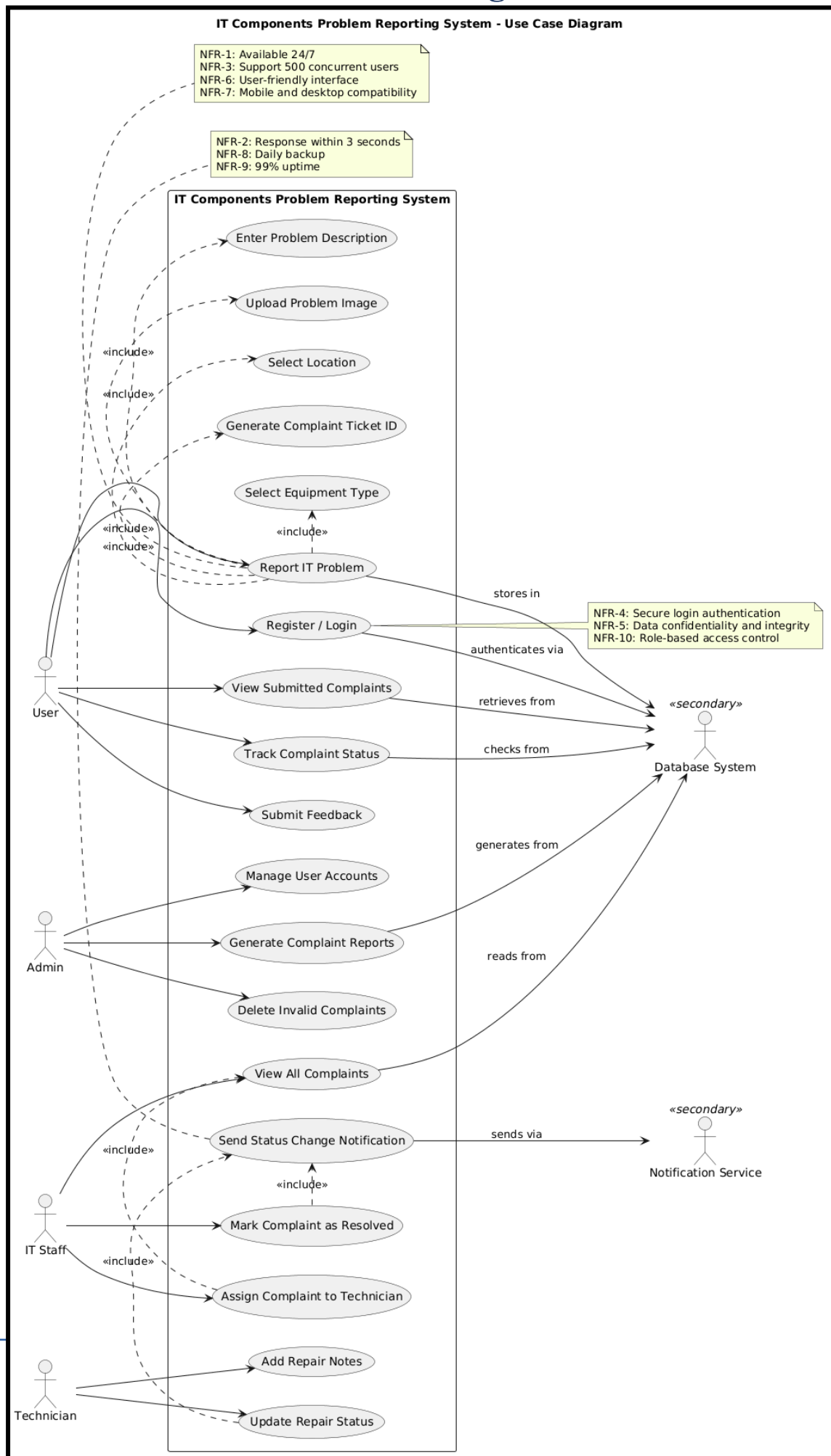
FR ID	Priority	Description	Actor	Status
FR-01	High	The system shall allow users to create an account.	All Users	Mandatory
FR-02	High	The system shall allow users to log in using username and password.	All Users	Mandatory
FR-03	High	The system shall allow users to report IT equipment problems.	User	Mandatory
FR-04	High	The system shall allow users to select equipment type (PC, printer, projector, etc.).	User	Mandatory
FR-05	High	The system shall allow users to enter problem description.	User	Mandatory
FR-06	Medium	The system shall allow users to upload images of the problem.	User	Mandatory
FR-07	High	The system shall allow users to select location (lab/office).	User	Mandatory
FR-08	High	The system shall generate a complaint ticket ID automatically.	System	Mandatory
FR-09	Medium	The system shall store all complaint information in the database.	User	Mandatory
FR-10	High	The system shall allow users to view previously submitted complaints.	User	Mandatory
FR-11	High	The system shall allow users to track the status of their complaints.	User	Mandatory
FR-12	High	The system shall allow IT staff to view all submitted complaints.	IT Staff	Mandatory
FR-13	High	The system shall allow IT staff to assign complaints to technicians.	IT Staff	Mandatory
FR-14	Medium	The system shall allow technicians to view complaints assigned to them.	Technician	Mandatory
FR-15	High	The system shall allow technicians to update repair status.	Technician	Mandatory
FR-16	Medium	The system shall allow technicians to add repair notes.	Admin	Mandatory
FR-17	High	The system shall allow IT staff to mark complaints as resolved.	System	Mandatory
FR-18	High	The system shall allow users to give feedback after problem resolution.	IT Staff	Mandatory
FR-19	Medium	The system shall allow administrators to manage user accounts.	User	Optional

FR-20	Low	The system shall allow administrators to generate complaint reports.	Admin	Mandatory
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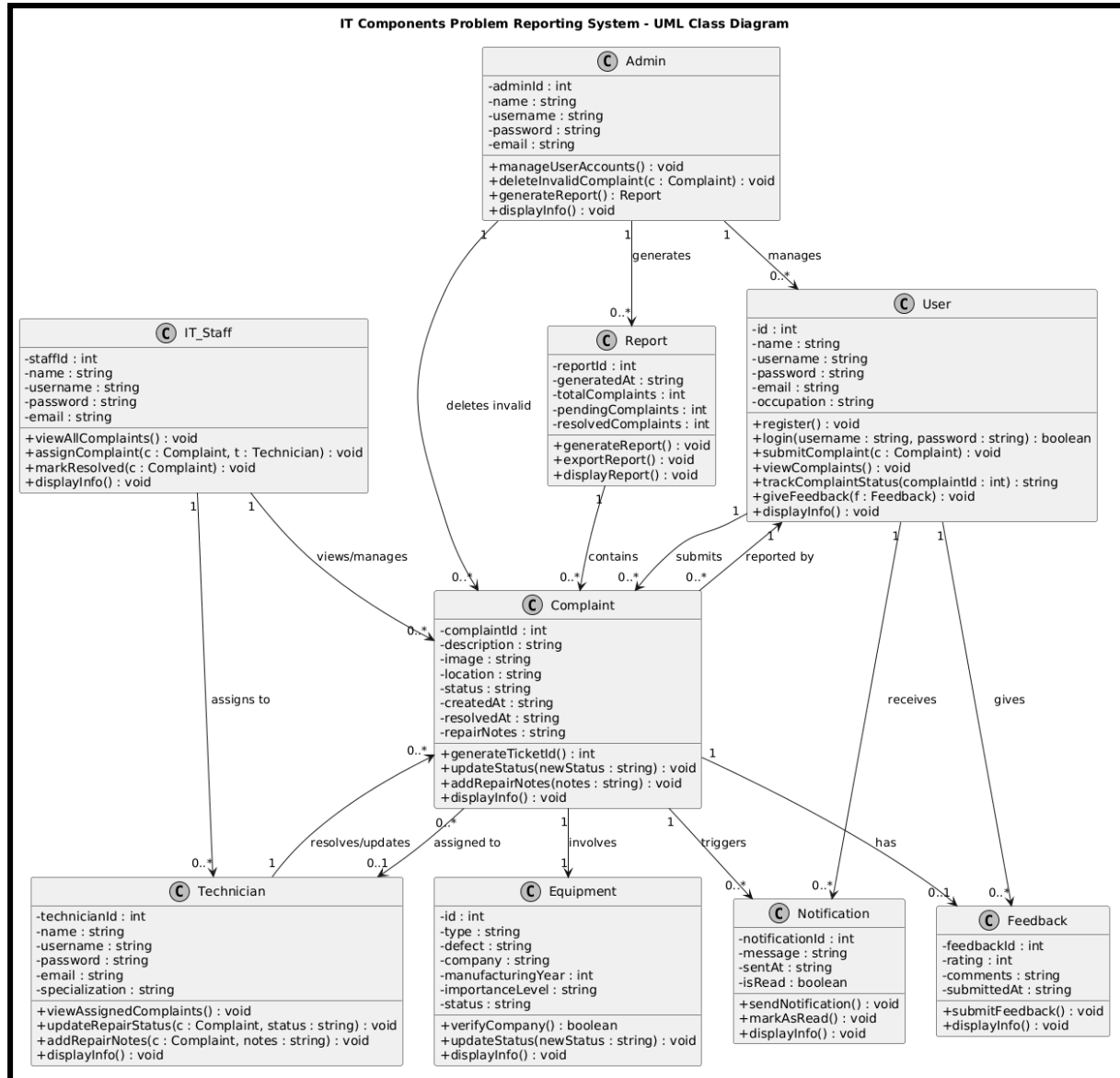
4. Non-Functional Requirements

NFR ID	Category	Requirement Description
NFR-01	Availability	The system shall be available 24/7 for users.
NFR-02	Performance	The system shall respond to user requests within 3 seconds.
NFR-03	Scalability	The system shall support at least 500 users simultaneously.
NFR-04	Security	The system shall ensure secure login authentication.
NFR-05	Security	The system shall ensure data confidentiality and integrity.
NFR-06	Usability	The system shall have a user-friendly interface.
NFR-07	Compatibility	The system shall be compatible with mobile and desktop devices.
NFR-08	Reliability	The system shall maintain backup of data daily.
NFR-09	Availability	The system shall ensure system uptime of 99%.
NFR-10	Access Control	The system shall follow proper access control for different users (Admin, IT Staff, User).

Use-Case Diagram



Class Diagram



UML -Specifications

Class	UML Specifications
Complaint	1. A User can register many complaints after which a complaint will be generated containing all the important information about the complaint. 2. Right after a complaint is registered a notification will be generated and can be viewed by everyone in the system. 3. That complaint can be resolved by the technicians but IT staff will assign the technician for a specific problem. 4. A complaint should always be about one Equipment.
User	1. User can register by their unique emails and user names 2. Many Users are managed by the Admin 3. Users can view the notification and can give feedback.
Admin	1. Admin generates many reports of complaints registered by the users. 2. If an admin finds any invalid complaint he/she can remove it. 3. Admin manages a User's information.
Report	1. The report is generated by the admin of the system and can contain the information and status about complaints.
IT_staff	1. IT staff and views and manages complaints generated by users and can assign many technicians for a specific task. For example if a project is not working IT_Staff will assign the repair duties to the respective person responsible. 2. IT Staff can view reports generated by the admin.
Technician	1. A technician can be assigned many tasks. 2. After resolving a task technician should know about the updated device information
Equipment	1. A complaint must be about and equipment 2. The information stored about the equipment must be updated in real time to know the exact status.
Notification	1. A notification can be viewed by users. 2. The notification is basically a string message. 3. Every notification must have its unique ID
Feedback	1. A User can submit a feedback about the complaint he or she registered. 2. A feedback about the general response of IT_staff to a complaint.

